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TREND-BASED MOCK PAPER

GATE Architecture and Planning

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Architecture and Planning
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Scale 1:100, drawing 10cm -> actual:

- (A) 1m
- (B) 10m
- (C) 100m
- (D) 1000m

Section B: Architecture and Planning

Q2. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \|u\| \|v\|$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q3. [1-NAT] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) $e^{-3/s}$

Q5. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q6. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q7. [1-NAT] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q9. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q10. [1-NAT] [1 mark]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q12. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a, b]$ and differentiable on (a, b)
- (D) Both continuous and differentiable everywhere

Q13. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q14. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q16. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q17. [2-NAT] [2 marks]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MSQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q19. [2-MCQ] [2 marks]

The Laplace transform of $t e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q20. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q21. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q22. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q23. [2-NAT] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q25. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q26. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q27. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| \cdot \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| \cdot \|v\|}$

Q28. [2-MSQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$

(D) $y = x^2$

Q29. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q31. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q32. [2-NAT] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MSQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q34. [1-MCQ] [1 mark]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q35. [1-MCQ] [1 mark]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q36. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q37. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q38. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q39. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q40. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q41. [2-NAT] [2 marks]

Camber on road provides:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q43. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q44. [1-MCQ] [1 mark]

Cantilever UDL w, L. Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$

(D) $wL^2/8$

Q45. [1-NAT] [1 mark]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-MCQ] [1 mark]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q47. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
- (B) 0.43
- (C) 0.5
- (D) 0.75

Q48. [2-NAT] [2 marks]

Cantilever UDL w , L . Fixed end moment:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-NAT] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle
- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q51. [2-MCQ] [2 marks]

Cantilever UDL w , L . Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$
- (D) $wL^2/8$

Q52. [1-NAT] [1 mark]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

Kath-Kuni from:

- (A) Rajasthan
- (B) Himachal Pradesh

- (C) Kerala
- (D) Gujarat

Q54. [1-NAT] [1 mark]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q56. [2-NAT] [2 marks]

M20 concrete mix ratio:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-NAT] [2 marks]

M20 concrete mix ratio:

(Numerical Answer Type — Enter the numerical value)

Q58. [2-NAT] [2 marks]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Trap in plumbing:

- (A) Prevents smell
- (B) Stores water
- (C) Both
- (D) None

Q60. [1-MCQ] [1 mark]

Sun path in northern hemisphere:

- (A) South
- (B) North
- (C) East
- (D) West

Q61. [1-MCQ] [1 mark]

M20 concrete ratio:

- (A) 1:1:2
- (B) 1:1.5:3
- (C) 1:2:4
- (D) 1:3:6

Q62. [1-NAT] [1 mark]

Zoning regulates:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Golden ratio approximately:

- (A) 1.414
- (B) 1.618
- (C) 2.0
- (D) 3.14

Q64. [1-MCQ] [1 mark]

Load-bearing wall thickness:

- (A) 23cm
- (B) 11cm
- (C) 46cm
- (D) 5cm

Q65. [1-MCQ] [1 mark]

Hardscape includes:

- (A) Plants
 - (B) Pavement
 - (C) Water
 - (D) Trees
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	B	1
3	1MCQ	B	1
4	1MCQ	B	1
5	1NAT	C	1
6	1MCQ	B	1
7	1MCQ	A	1
8	1MCQ	C	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1NAT	B	1
13	1MCQ	A	1

14	1MCQ	B	1
15	1NAT	A	1
16	1MCQ	C	1
17	1MCQ	B	1
18	1MCQ	A	1
19	1MCQ	A	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	A	1
23	1MCQ	B	1
24	1NAT	C	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1NAT	A	1
28	1MCQ	B	1

29	1MCQ	C	1
30	1MCQ	A	1
31	2NAT	B	2
32	2MSQ	C	2
33	2MCQ	A	2
34	2NAT	C	2
35	2MSQ	A	2
36	2MCQ	A	2
37	2MCQ	C	2
38	2MCQ	B	2
39	2MCQ	A	2
40	2NAT	A	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2MCQ	A	2

44	2NAT	A	2
45	2MCQ	A	2
46	2MCQ	A	2
47	2MSQ	A	2
48	2MCQ	B	2
49	2NAT	A	2
50	2NAT	A	2
51	2MCQ	B	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2MCQ	A	2
55	2MCQ	B	2
56	2NAT	A	2
57	2NAT	B	2
58	2NAT	B	2

59	2NAT	A	2
60	2NAT	A	2
61	2MCQ	B	2
62	2MCQ	A	2
63	2MCQ	C	2
64	2MCQ	A	2
65	2MSQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: History | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: Vernacular | Difficulty: easy

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Building Materials | Topic: Concrete | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Urban Planning | Topic: Zoning | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Design | Topic: Principles | Difficulty: difficult

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Technology | Topic: Structures | Difficulty: difficult

Q8.

Answer: C

Solution:

Correct Answer: C

Subject: Climate & Services | Topic: Plumbing | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Climate & Services | Topic: Solar | Difficulty: difficult

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Landscape Architecture | Topic: Elements | Difficulty: moderate

Q11.

Answer: A

Solution:

$$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa.}$$

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q12.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q13.

Answer: A

Solution:

$$FEM = wL \cdot (L/2) = wL^2/2 \text{ (hogging).}$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q14.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q15.

Answer: A

Solution:

$$DF_{AB} = 4/(4+3) = 4/7 = 0.57.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q16.

Answer: C

Solution:

$$R_A = (L-x)/L. \text{ Linear decreasing from } L \text{ to } 0.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q17.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q18.

Answer: A

Solution:

$$|<u,v>| \leq \|u\| \cdot \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q20.

Answer: B

Solution:

$$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0. \text{ Roots at } x=1 \text{ (double?) and } x=-2. f(x)=(x-1)^2(x+2). \text{ Two distinct real roots.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q21.

Answer: A

Solution:

$$L\{e^{(-at)}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = 1+4+9 = 14.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q31.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q32.

Answer: C

Solution:

$y = A*\cos(2x) + B*\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$L\{tf(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q43.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q44.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q45.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

$E_c = 5000 * \sqrt{f_{ck}} = 5000 * 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q47.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x^m$ and $y(1) = 0$ gives $y = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q48.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q49.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q50.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: B

Solution:

$\text{proj} = \frac{(v \cdot w)}{(w \cdot w)} * w = \frac{4}{5} * (2, 1, 0) = (\frac{8}{5}, \frac{4}{5}, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: A

Solution:

$DF_{AB} = \frac{4}{4+3} = \frac{4}{7} = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q53.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q54.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x^m$ and $y(1) = 0$ gives $y = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q55.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q56.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: difficult

Q57.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q58.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q59.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q60.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q61.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q62.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000$ MPa.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q63.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q64.

Answer: A

Solution:

$FEM = wL \cdot (L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q65.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 52 marks
Civil Engineering: 10 questions, 17 marks
Structural Analysis: 8 questions, 13 marks
Architecture: 7 questions, 10 marks
Climate & Services: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Building Materials: 1 questions, 1 marks
Urban Planning: 1 questions, 1 marks
Design: 1 questions, 1 marks
Construction Technology: 1 questions, 1 marks
Landscape Architecture: 1 questions, 1 marks

Difficulty Distribution:

Easy: 14 | Moderate: 40 | Difficult: 11

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.